Executive Summary

This analysis examined over 10,000 collision records from Toronto's open data portal to identify the factors that most strongly predict severe injury outcomes. Logistic regression models revealed that vulnerable road users—particularly pedestrians, cyclists, moped drivers, and motorcyclists—face **13 to 34 times higher odds** of severe injury compared to drivers. Older adults (55+) also show a significantly increased risk, with the predicted probability of severe injury rising from **74% among 25-year-olds to 83% among 75-year-olds**. Additionally, the 20–24 age group is identified as a high-risk group.

Environmental and temporal patterns indicate that **dusk (artificial) lighting** and **morning and noon rush hours** are the periods of highest risk. The months of **January, October, November, and December** also show elevated risk.

These insights support targeted interventions, including improved lighting and protected infrastructure, senior-friendly crossing designs, rush-hour enforcement, winter safety campaigns, school sensitization programs, and social media awareness campaigns. Collectively, these measures align with **Toronto's Vision Zero objective** to eliminate severe injuries and fatalities.

All results are based on **odds ratios and predicted probabilities** from logistic regression models built in Python (JupyterLab). Detailed statistical outputs are summarized in the appendices.

Definition of Business Problem

Traffic collisions resulting in severe injuries or fatalities continue to pose a significant challenge to public safety in Toronto. Annually, the city records thousands of these incidents. This disproportionately affects vulnerable road users such as pedestrians, cyclists, and older adults, all of which are at elevated risk experiencing a serious injury.

This study aims to examine the user, environmental conditions, and temporal variables to determine which factors most strongly correlate with the severity of injuries sustained in traffic collisions. By identifying these predictors, the research seeks to inform targeted interventions that enhance roadway safety.

Findings from this analysis will provide empirical support for municipal decision makers (including transportation planners and public safety officials) enabling them to prioritize infrastructure upgrades, enforcement strategies and policy initiative. Ultimately, the goal is to utilize a data-driven approach to eliminating severe traffic injuries and fatalities across the city.

Description of Data

The analysis uses the *Toronto Motor Vehicle Collisions Involving Killed or Seriously Injured Persons* dataset from the City of Toronto Open Data Portal (2006–2023).

- **Data Source:** Toronto Open Data Portal
- **Dataset Size:** *(10044, 169)*

- **Columns Used:** *UserInvType_*, UserAgeRange_*, EnvLightingCondition_*, EnvRoadSurfaceCondition_*, TimePeakOffPeak_*, TimeSeason_*, TimeMonth_*,TimeRange_**

Data Cleaning and Preparation:

A series of steps were performed to prepare the data for the Logistical Regression:

- Renamed columns to more meaningful names
- Identified and removed records with missing critical data
- Converted date and time into useable numeric features
- Created dummy variables for all categories: *Driver (Involvement Type), Age 20–24 (Age Group), Daylight (Lighting Condition), Dry (Road Surface), 2006 (Year), January (Month), 12 (Hour), 12:00–14:59 (Time Range), Peak (Peakoff - Peak), Non - Winter (Season)*
- Cleaned and standardized categorical predictor variables

Proposed Analysis

A *Logistic Regression Model* was used to estimate the likelihood that a collision results in a *severe injury* (Fatal or Major), given different user, environmental, and temporal conditions.

Logistic regression is appropriate for this analysis because the outcome variable (*Severity of Injury*) is binary (severe vs. non-severe). Each variable was assessed using:

- p-values for statistical significance,
- Confidence Intervals (CI) for reliability
- Odds Ratios / Percentage Change to interpret the direction and magnitude of impact.
- Predicted Probabilities which communicated risk

Multiple models were generated in Python (JupyterLab) using the statsmodels library for each slice.

Interpreting the Results

Four statistical measures are presented throughout this report:

1. P-values: P-values indicate whether the effect of a variable is statistically meaningful or likely due to random chance.
2. Confidence Interval: Confidence intervals show the range within which the true effect is likely to fall, providing a measure of reliability and precision.
3. Odds Ratios (OR): show how much more or less likely a collision outcome is compared with a baseline group. For example, an OR = 2 means the odds are twice as high as the baseline.
4. Predicted Probabilities: express those relationships as actual chances (percentages) of a severe injury occurring.

Odds Ratios are used to describe relative risk across categories (e.g., involvement type, lighting) while Predicted Probabilities are used to communicate the absolute likelihood of a severe injury for each group.

User Questions and Answers

1. **How much higher are the odds of severe injury for older adults compared to younger drivers?**

   Older adults are substantially more vulnerable to severe injuries in collisions compared to younger drivers. Drivers aged 55- 69 have 40- 80% higher odds of experiencing a severe injury than those aged 20- 24. This risk increases sharply with age. Individuals aged 70- 79 are about twice as likely at 100% - 120%, and those aged 85- 89 are over three times as likely to sustain severe injuries. These results are derived from the odds ratios in Model B (Appendix A), where all p-values indicate strong statistical significance. This confirms that age is a reliable and meaningful predictor of injury severity.

This trend highlights the need for senior-focused safety interventions, such as improved lighting, longer pedestrian crossing times, and speed management in areas with a high concentration of older adults.

2. **Which involvement types significantly increase the odds of a severe injury?**

   Unprotected Road Users face the highest likelihood of severe injury. The analysis shows that cyclists, motorcyclists, moped drivers, and pedestrians are between 13 and 34 times more likely to suffer a severe injury than drivers. In contrast, truck drivers and passengers experience notably lower odds which is likely due to vehicle protection. According to the odds ratios reported in Model A (Appendix A), these involvement types show highly significant results, with confidence intervals entirely above 1.0, a clear indication of a strong positive impact on injury severity.

   These findings suggest that investments in protected cycling lanes, dedicated motorcycle infrastructure, and pedestrian-priority zones can significantly reduce injury severity across Toronto's streets.

3. **Is the effect of age statistically significant in predicting severe injuries?**

   Yes. Age is a statistically significant factor in predicting the likelihood of severe injuries. Individuals aged 50- 94 consistently show higher odds of severe injury compared to younger drivers. For these age groups, the odds increase by approximately 1.3x to 11.7x, confirming that injury severity rises sharply as people age. The pattern is confirmed by Model B (Appendix A), where all associated p-values demonstrate strong statistical significance, reinforcing age as a key predictor of severe injury outcomes. This demonstrates the importance of age-sensitive transportation planning, including driver assessment programs, mobility alternatives for seniors, and public education on

safe driving among older populations.

4. **Does being a pedestrian increase or decrease the odds of severe injury relative to drivers?**

   Being a pedestrian greatly increases the risk of severe injury compared to being a driver. Pedestrians are approximately 13 times more likely to suffer a severe injury when involved in a collision. This is both statistically and practically significant. Model A (Appendix A) reports a pedestrian odds ratio of 13.76 with a 95 % confidence interval (11.6 – 16.3) and a highly significant p-value, confirming that pedestrians face a statistically strong and substantially higher risk of severe injury compared with drivers. This reinforces the importance of pedestrian safety infrastructure such as, raised crosswalks, refuge islands, better intersection lighting, and reduced vehicle speeds in pedestrian-heavy zones.

5. **What is the predicted probability that a collision involving a 75- 79 years old results in severe injury?**

   The model predicts that a collision involving a person aged 75–79 has an 83% probability of resulting in a severe injury. This probability is based on the predictive outputs from Model A (Appendix A), which translate the statistical model into an estimated real-world likelihood of severe injury.

   This underscores the vulnerability of older adults in traffic collisions and highlights the need for policies and design improvements that minimize collision impact, including lower urban speed limits and better access to emergency medical response services.

6. **By how many percentage points is the probability of a severe injury increase from age 25-29 to age 75-79**

The probability of a severe injury rises from 74% for those aged 25- 29 to 83% for those aged 75- 79, an 8.5 percentage-point increase. While this may appear modest, it represents a meaningful escalation in real-world risk. This comparison is drawn from the predicted-probability results in Model A (Appendix A), which express the difference in absolute risk between the two age groups.

This finding emphasizes the importance of early safety interventions for aging drivers, such as defensive-driving refreshers and community mobility programs to reduce collision exposure.

Charts

Severity risk increases steadily with age. Individuals aged 75+ face an 80- 99% probability of severe injury, emphasizing the need for senior-focused safety measures:

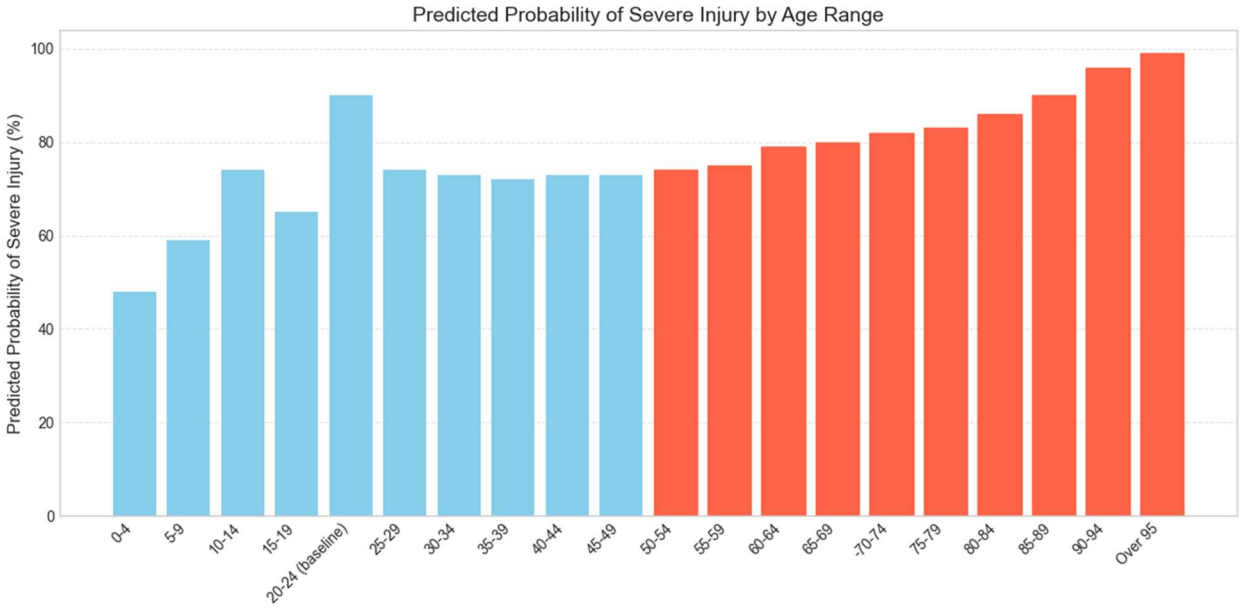


Figure 1- Predicted Probability of Severe Injury by Age Range

Vulnerable users- cyclists, motorcyclists, moped drivers, and pedestrians are 13- 34x more likely to be severely injured than drivers. Passengers and truck drivers face lower risks:

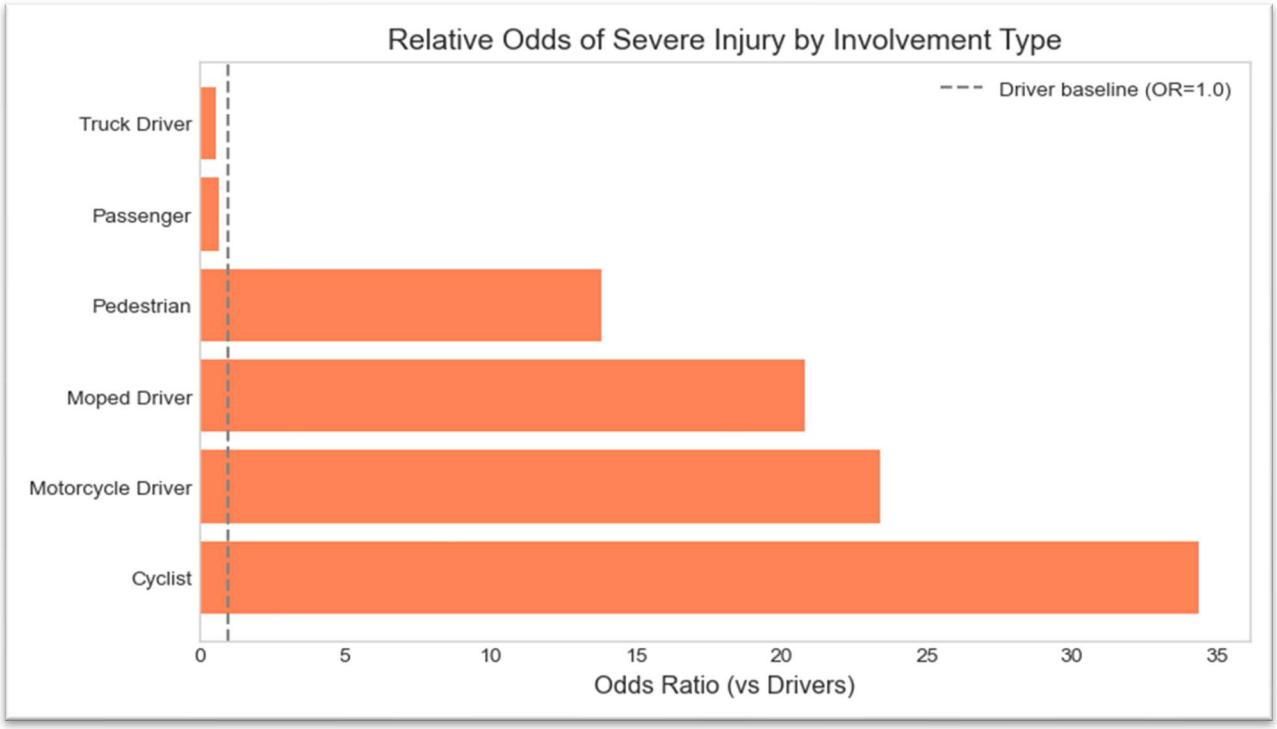


Figure 2- Relative Odds of Severe Injury by Involvement Type

Managerial Insights (User Factors)

The results make it clear that age and involvement type are critical determinants of injury severity. Older adults, pedestrians, cyclists, moped, and motorcyclists face the greatest danger, while enclosed-vehicle occupants are far better protected.

To address these risks, decision-makers should focus on safer infrastructure, targeted education campaigns, and enhanced medical readiness to protect Toronto's most vulnerable road users.

Environment Questions and Answers

1. **How much higher are the odds of a severe collision under Dusk lighting compared to Daylight conditions?**

   Collisions that occur under dusk artificial lighting are almost twice as likely to result in a severe injury compared to those occurring in daylight. This suggests that visibility challenges at dusk contribute significantly to crash severity. As shown in Model A

(Appendix B), dusk with artificial lighting has an odds ratio of 1.9 (95 % CI 1.1 - 3.1), indicating a statistically significant increase in severity compared to daylight. In contrast, dusk under natural light is not statistically significant (CI 0.7- 1.7).

The city can mitigate this risk through better street lighting, enhanced visibility measures, and driver-awareness programs during early evening hours.

2. **Which specific road surface conditions are statistically significant predictors of severe outcomes when compared to Dry roads?**

   Two surface conditions stand out:

   - *"Other" road surfaces are associated with roughly twice the odds of severe injury compared to dry conditions.*
   - *"Ice" surfaces show a 58% lower likelihood of severe injury, likely due to reduced vehicle speeds.*

   Model B (Appendix B) shows that "Other" road surfaces are associated with roughly twice the odds of severe injury (95 % CI 1.1 – 3.9), while "Ice" surfaces are linked to significantly lower odds (95 % CI 0.2 – 0.7), reflecting slower driving speeds and reduced impact forces.

   These findings indicate that while icy conditions slow traffic and reduce impact severity. It is still difficult to identify and address road surface conditions that are classified as others since this seems to be a generic classification.

3. **Are there differences in severity between Dark and Daylight conditions statistically significant or can they be due to random variation?**

   Yes, the differences in severity between Dark and Daylight conditions statistically significant. Collisions in natural darkness (without artificial lighting) are significantly more likely to be severe, while those under dark (with artificial lighting) lighting show no

statistically meaningful difference from daylight conditions and can be explained by a random variation. Model A (Appendix B) indicates that natural darkness reduces the odds of severe injury by approximately 11 % (95 % CI 0.79- 0.99), a statistically significant effect. Collisions under artificial lighting show a smaller, non-significant difference (95 % CI 0.8 – 1.1).

Improving illumination and reflective road markings in dark areas can reduce these severity outcomes.

4. **Does the change in reduced visibility (dark to dusk to daylight) show a decrease or increase in odds of a severe collision?**

   The analysis shows that visibility does influence injury severity, but not uniformly. Collisions in *natural darkness* have *11% lower odds* of being severe, while *natural dusk* has a slight, not statistically significant *12% increase. As detailed in Model A (Appendix B), collisions in natural darkness have about 11 % lower odds of being severe (95 % CI 0.79 - 0.999), while natural dusk shows a small, non-significant 12 % increase (95 % CI 0.74 - 1.70).*

   Although not all visibility effects are statistically significant, they point to the importance of visibility management in safety planning- better lighting, reflective infrastructure, and driver awareness.

5. **How do the predicted probabilities of severe conditions differ between packed snow and ice surfaces?**

   Collisions on packed snow are significantly more likely to result in severe injury (78% probability) than those on ice (54% probability). This suggests that even though drivers slow down on ice, packed snow may create a deceptive sense of safer driving conditions for road users. Based on predicted probabilities from Model B (Appendix B), collisions on

packed snow have an estimated 78 % chance of severe injury versus 54 % on ice, underscoring the risk of deceptive traction on snow-covered surfaces.
Snow-removal operations and winter tire awareness campaigns should focus on mitigating risks from snow-covered surfaces.

6. **What overall pattern emerges between lighting condition and severity risk? Do probabilities rise as lighting decreases?**
   The results show that injury severity does not increase steadily as lighting decreases (dawn -> dusk -> dark). Instead, artificially lit dusk conditions are the most dangerous, while natural-light conditions are generally safer. These results are confirmed by the predicted probabilities in Model B (Appendix B), showing that artificial dusk conditions produce the highest severity likelihood among all lighting scenarios.
   This indicates that transition periods between light and dark are particularly risky and should be prioritized for visibility enhancements and speed enforcement.

7. **What is the most dangerous combination of lighting conditions and road surface conditions?**
   The analysis did not include interaction effects between lighting and road-surface conditions, so this combination could not be evaluated within scope. Future research using more advanced modeling tools (i.e. other libraries) could help identify these high-risk intersections.

CHARTS

Dusk with artificial lighting nearly doubles the odds of a severe injury, suggesting visibility challenges during transition hours:

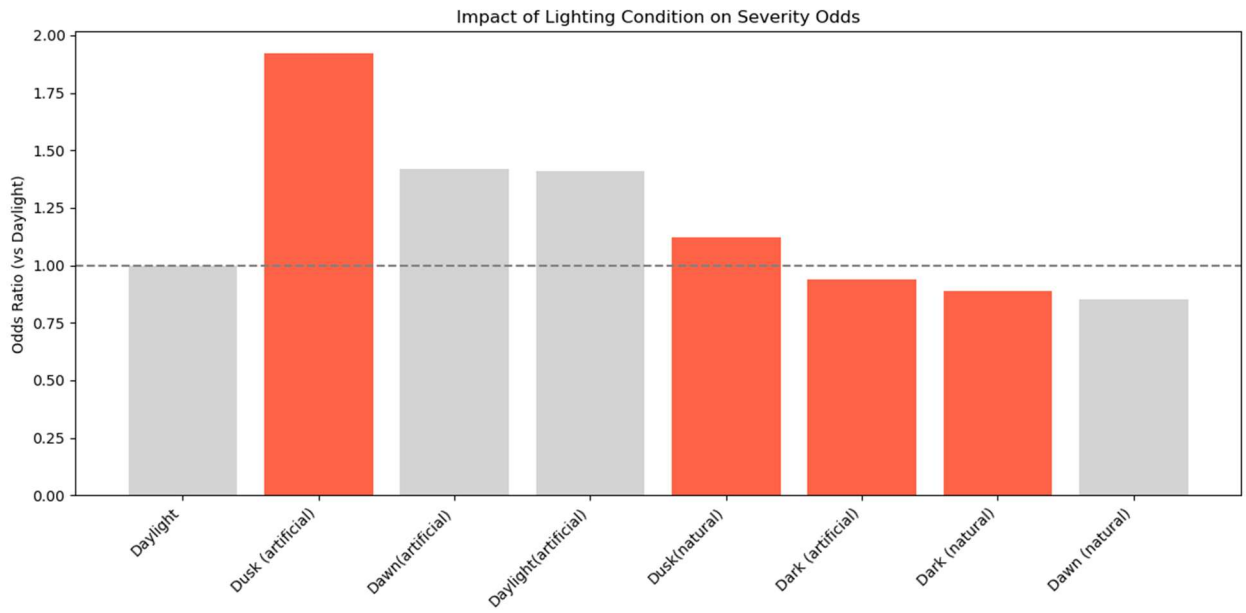


Figure 3- Impact of Lighting Condition on Severity Odds (grey bars indicate not statistically important)

Collisions on packed snow and "other" surfaces show higher severity risks. Ice conditions show fewer severe outcomes, likely due to slower driving speeds:

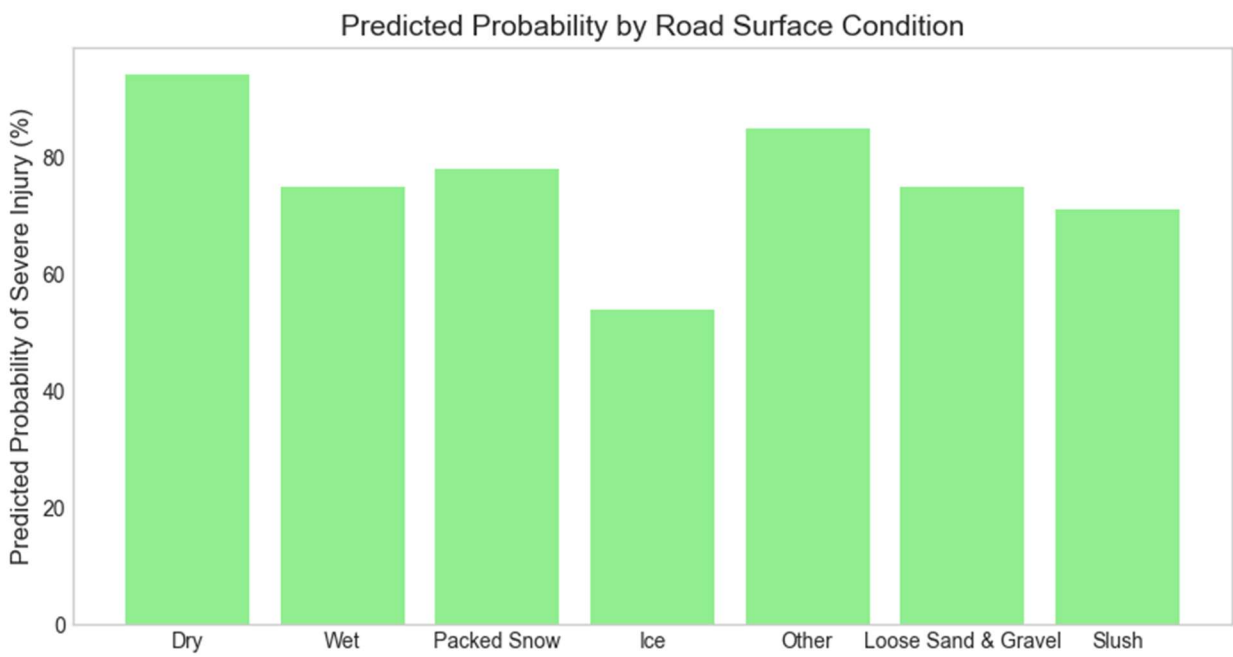


Figure 4- Predicted Probability by Road Surface Condition

Managerial Insights (Environmental Factors)

Environmental conditions have a measurable influence on injury severity, particularly lighting at dusk and non-standard road surfaces.

The findings suggest targeted actions:

- Invest in street lighting upgrades in high-traffic dusk corridors.
- Maintain consistent winter road-surface treatment.
- Implement seasonal safety campaigns during fall and winter to promote safe driving behavior under variable conditions.

Time Questions and Answers

1. **What is the predicted probability of a severe collision during winter compared to not-winter**

   Collisions during winter have a 73.2% probability of being severe, compared to 73.9% in non-winter months. This small difference (0.7%) suggests that season alone does not significantly affect injury severity. This comparison is drawn from predicted probabilities in Model B (Appendix C) and the baseline intercept values in Appendix D, confirming that winter conditions have no statistically meaningful effect on severity.

   Drivers appear to adapt their behavior to winter conditions, indicating that driver education and preparedness campaigns should continue to reinforce safe winter habits rather than focus solely on the season itself.

2. **What months have a higher severity probability?**

The months of October, November, and December show the highest probabilities of severe collisions, with November being the peak month (78% predicted probability). The safest month is April, at roughly 69%. This pattern suggests that shorter daylight hours and poorer driving conditions late in the year contribute to higher risk. Predicted probabilities from Model C (Appendix C) highlight the October–December months as periods of elevated severity risk, consistent with shorter daylight and adverse weather conditions.

The city should therefore plan fall and early-winter safety campaigns emphasizing reduced speed, increased alertness, and tire preparedness.

3. **How do predicted probabilities vary across the time-of-day ranges?**

   Severity risk peaks during morning rush hours (6:00 AM- 8:59 AM), with a predicted probability of 78% for severe injury. The lowest-risk period is early morning (3:00 AM-5:59 AM) at 69%. This pattern reflects greater traffic volume and speed variation during commute hours. Model E (Appendix C) shows a clear morning-peak effect, with the 6 – 8:59 AM time range exhibiting the highest predicted probability of severe injury. Targeted enforcement and driver-awareness initiatives during peak periods can help reduce collision severity.

4. **How do the odds of severe traffic injuries during off-peak hours compare to those of peak hours**

   The odds of severe injury are 17% lower during off-peak hours compared to peak hours. According to Model A (Appendix C) and baseline values from Appendix D, off-peak hours show approximately 17 % lower odds of severe injury, a statistically significant difference.

This statistically significant difference highlights the importance of managing traffic congestion and speed behavior during rush-hour windows.

5. **Is season a significant risk factor influencing the likelihood of severe collisions?**

   No. The analysis shows that season alone is not a significant risk factor. The odds of severe injury are slightly lower (- 3.6%) in winter, which is not statistically significant. Model B (Appendix C) reports an odds ratio of 0.96 (95 % CI 0.86 - 1.07) with a non-significant p-value, providing little evidence that winter increases the odds of severe injury.

   Winter safety efforts should therefore complement, rather than replace, broader initiatives focused on road design and driver behavior.

6. **Does time of day (peak vs. off peak) have a statistically significant impact on severity?**

   Yes. Collisions during off-peak hours have 17% lower odds of getting a severe injury. As shown in Model A (Appendix C), off-peak collisions have an odds ratio below 1 (95 % CI 0.75 – 0.90), with strong statistical evidence that time of day significantly affects injury severity.

   This confirms that peak-hour congestion and higher interaction rates drive increased severity levels.

7. **In which direction does off-peak affect accident severity (increase or decrease the likelihood of severe injuries)**

   The effect is negative. Off-peak conditions reduce the likelihood of severe injuries by approximately 17%. Model A (Appendix C) confirms a statistically significant negative

relationship (95 % CI 0.75 – 0.90), meaning that off-peak periods are associated with fewer severe outcomes compared to peak hours.

This emphasizes that traffic management and enforcement during peak hours can directly reduce severe outcomes.

CHARTS

Collision severity increases toward year-end (October–December) into the early part of the year, likely due to reduced daylight and changing weather:

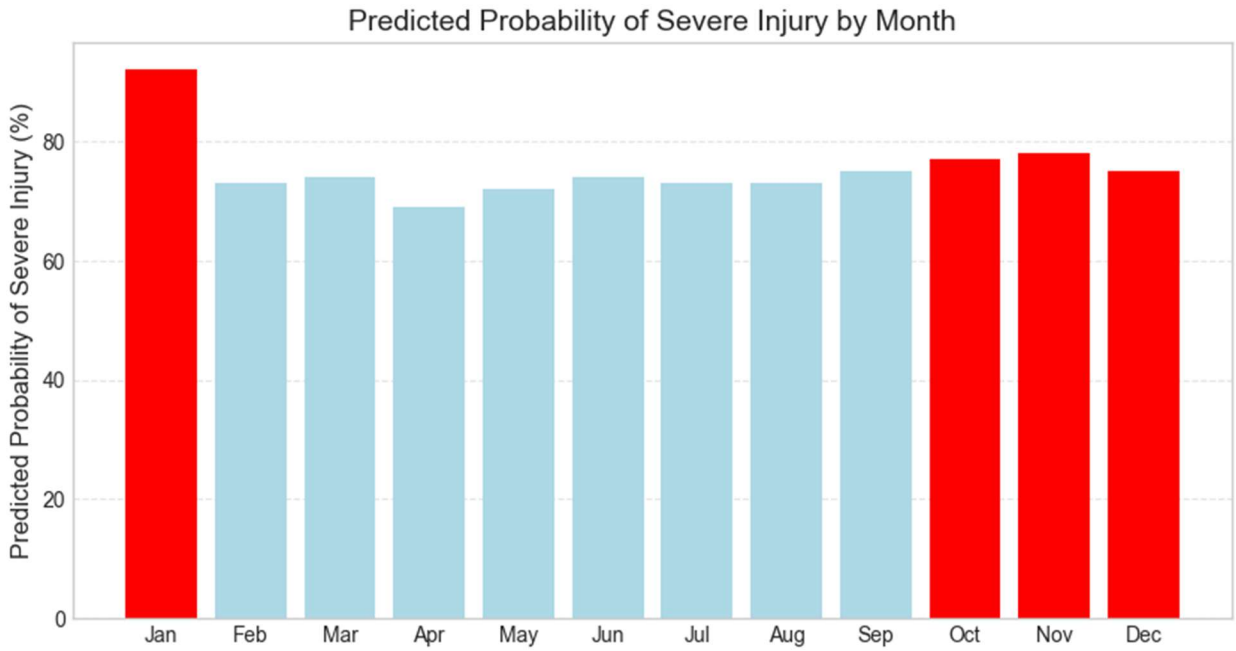


Figure 5- Predicted Probability of Severe Injury by Month

Morning rush hours (6–8:59 AM) along with (12:00-2:59PM) have the highest predicted probability of severe injuries. Enforcement and awareness should focus on peak commuting times:

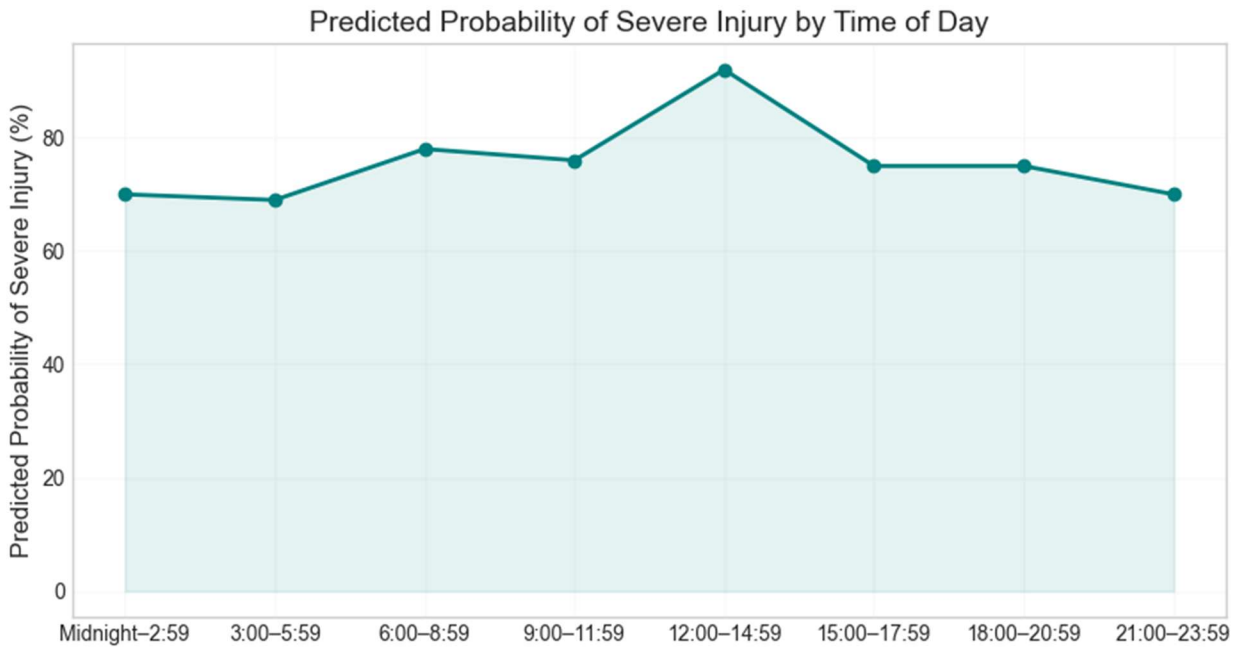
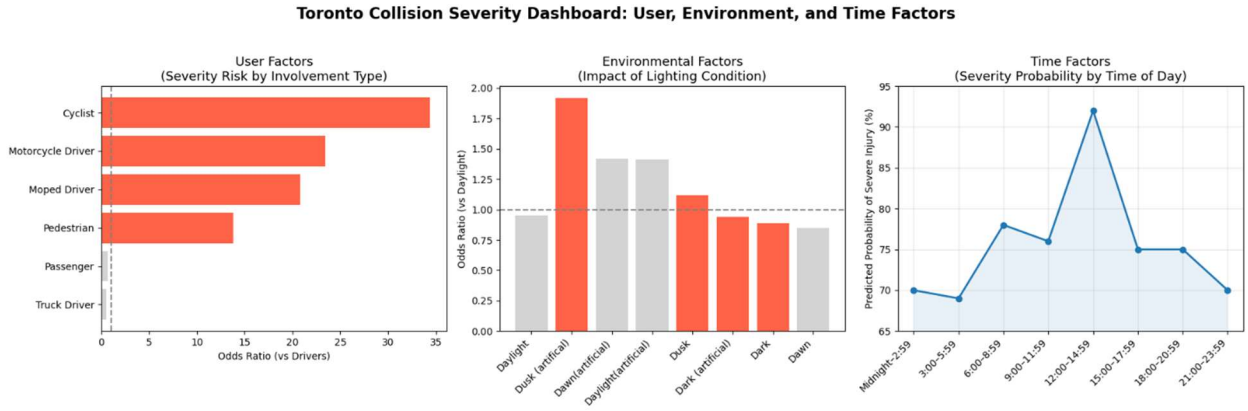


Figure 6- Predicted Probability of Severe Injury by Time of Day

Managerial Insights (Time Factors)

Timing plays a clear role in collision severity. Rush hours, particularly morning commutes (6 AM-8:59AM), show the greatest risk, while seasonal differences are minimal. Municipal leaders should prioritize speed management and enforcement during peak hours, seasonal awareness campaigns in late fall and improved signal timing in commuter corridors to reduce high-severity incidents.

CONCLUSION



Logistic regression analysis demonstrates that vulnerable road users (VRUs)—specifically pedestrians, cyclists, and motorized two-wheeler riders—experience 13 to 34 times higher odds of severe injury than vehicle occupants. Age is a critical determinant of outcome severity; the predicted probability of severe injury increases from 74% at age 25 to 83% at age 75, with the 20–24 age cohort also identified as a high-risk demographic. Furthermore, temporal and environmental factors significantly influence risk levels, with peak danger occurring during morning and midday rush hours, under artificial dusk lighting, and during the winter months (October through January).

Data and Code Availability

All code and relevant files are stored here: https://drive.google.com/drive/folders/1t6MoUdou-3VU8HN0dL1emab5bXMtExD1?usp=drive_link

The folder structure reflects the phases of the project. Please review the “ReadMe” in the location for a further breakdown. A high-level breakdown is as follows:

- Cleaned dataset (TorontoCity_traffic_collisions_KPI_Modified.csv)
- Python (JupyterLab) code notebooks used for data wrangling, making slices and logistic regression modeling
- .csv files are provided

- .xlsx files contain the manual analysis out the output files

APPENDIX A

All questions under User were answered using Appendix A. *Baseline: Driver (Involvement Type), Age 20–24 (Age Group)*

Model A [User Involvement] (Odds Ratio, Confidence Interval, P-values)

	Odds Ratio	**2.50%**	**97.50%**	**pval_new**	**Consideration**	***Odds Inc/Dec***
UserInvType_Wheelchair	1268277000.00000	0	inf	0.99833070000000000	Little Evidence	1.26828E+11
UserInvType_In_Line_Skater	1234460000.00000	0	inf	0.99918030000000000	Little Evidence	1.23446E+11
UserInvType_Moped_Passenger	206225600.00000	0	inf	0.99908300000000000	Little Evidence	20622559900
UserInvType_Cyclist	34.40779	21.173987	55.912758	0.00000000000000000	Very Strong	3340.779
UserInvType_Motorcycle_Driver	23.44595	15.249722	36.047371	0.00000000000000000	Very Strong	2244.595

UserInvType_Moped_Driver	20.81081	2.828074	153.139514	0.00287445900000000	Very Strong	1981.081
UserInvType_Pedestrian	13.75895	11.595644	16.325858	0.00000000000000000	Very Strong	1275.895
UserInvType_Motorcycle_Passenger	1.73423	0.87873	3.42263	0.11245330000000000	Little Evidence	73.4234
UserInvType_Cyclist_Passenger	1.48649	0.134653	16.409909	0.74629010000000000	Little Evidence	48.6486
Intercept	1.34546	1.25476	1.442704	0.00000000000000008	Very Strong	34.5455
UserInvType_Passenger	0.66595	0.594188	0.74637	0.00000000000277243	Very Strong	-33.40541
UserInvType_Truck_Driver	0.53679	0.322583	0.893228	0.01664093000000000	Strong	-46.32132
UserInvType_Other	0.29730	0.057594	1.534624	0.14746940000000000	Little Evidence	-70.27027

Model A [User Involvement] (Predictive Probability):

Involvement	PredProb_Severe
UserInvType_Wheelchair	1.00E+00
UserInvType_In_Line_Skater	1.00E+00
UserInvType_Moped_Passenger	1.00E+00
UserInvType_Cyclist	9.79E-01
UserInvType_Motorcycle_Driver	9.69E-01
UserInvType_Moped_Driver	9.66E-01
UserInvType_Pedestrian	9.49E-01
UserInvType_Motorcycle_Passenger	7.00E-01
UserInvType_Cyclist_Passenger	6.67E-01
UserInvType_Passenger	4.73E-01
UserInvType_Truck_Driver	4.19E-01
UserInvType_Other	2.86E-01
UserInvType_Driver___Not_Hit	3.60E-09

Model B [User Age Range] (Odds Ratio, Confidence Interval, P-values)

	Odds Ratio	2.50%	97.50%	pval_new	Considerations	*Odds Inc/Dec*
UserAgeRange_Over_95	196.777112	0.000227	170620260.2	0.4489491000	Little Evidence	19577.7112
UserAgeRange_90_to_94	11.73566	2.843073	48.44255363	0.0006629349	Very Strong	1073.566
UserAgeRange_85_to_89	4.0861	2.503719	6.668567211	0.0000000178	Very Strong	308.61
UserAgeRange_80_to_84	2.757026	1.906119	3.987784766	0.0000000722	Very Strong	175.7026
Intercept	2.21547	1.956899	2.508205796	0.0000000000	Very Strong	121.547
UserAgeRange_75_to_79	2.186817	1.605978	2.977729875	0.0000006779	Very Strong	118.6817
UserAgeRange_70_to_74	2.015279	1.514457	2.681719151	0.0000015295	Very Strong	101.5279
UserAgeRange_65_to_69	1.780955	1.372841	2.310392088	0.0000138471	Very Strong	78.0955
UserAgeRange_60_to_64	1.678282	1.314427	2.142855991	0.0000328461	Very Strong	67.8282
UserAgeRange_55_to_59	1.384763	1.115269	1.71937889	0.0031993940	Very Strong	38.4763
UserAgeRange_50_to_54	1.318292	1.072219	1.620836537	0.0087552880	Very Strong	31.8292
UserAgeRange_25_to_29	1.310161	1.087354	1.578621889	0.0045036520	Very Strong	31.0161
UserAgeRange_10_to_14	1.272868	0.900883	1.798448351	0.1712816000	Little Evidence	27.2868

UserAgeRange_40_to_44	1.222763	0.993219	1.505357732	0.0579834500	Moderate	22.2763
UserAgeRange_30_to_34	1.20236	0.988962	1.461805576	0.0645117800	Moderate	20.236
UserAgeRange_45_to_49	1.193575	0.970771	1.467516726	0.0932372800	Moderate	19.3575
UserAgeRange_35_to_39	1.165698	0.95476	1.423238925	0.1322195000	Little Evidence	16.5698
UserAgeRange_15_to_19	0.853051	0.694391	1.047962191	0.1300842000	Little Evidence	-14.6949
UserAgeRange_5_to_9	0.64725	0.446158	0.938976738	0.0219255800	Strong	-35.275
UserAgeRange_unknown	0.53056	0.364888	0.771450668	0.0009048383	Very Strong	-46.944

Model B [User Age Range] (Predictive Probability):

AgeRange	**PredProb_Severe**
UserAgeRange_Over_95	0.9977110
UserAgeRange_90_to_94	0.9629630
UserAgeRange_85_to_89	0.9005240
UserAgeRange_80_to_84	0.8593160
UserAgeRange_75_to_79	0.8289090
UserAgeRange_70_to_74	0.8170100
UserAgeRange_65_to_69	0.7978020

UserAgeRange_60_to_64	0.7880540
UserAgeRange_55_to_59	0.7541730
UserAgeRange_50_to_54	0.7449390
UserAgeRange_25_to_29	0.7437620
UserAgeRange_10_to_14	0.7382200
UserAgeRange_40_to_44	0.7303850
UserAgeRange_30_to_34	0.7270590
UserAgeRange_45_to_49	0.7256010
UserAgeRange_35_to_39	0.7208710
UserAgeRange_15_to_19	0.6539680
UserAgeRange_5_to_9	0.5891470
UserAgeRange_unknown	0.5403230
UserAgeRange_0_to_4	0.4766360

APPENDIX B

All questions under Environment were answered using Appendix B. *Baseline Daylight (lighting condition) and Dry (Road surface).*

Model A [Light Conditions] (Odds Ratio, Confidence Interval, P-values)

	Odds Ratio	**2.50%**	**97.50%**	**pval_new**	**Consideration**	***Odds Inc/Dec***
EnvLightingCondition_Other	41592.67003	0.0000000000	7.436975E+212	0.9653254000	Little Evidence	4159167.003
Intercept	2.878201	2.7109920000	3.0557220000	0.0000000000	Very Strong	187.8201
EnvLightingCondition_Dusk__artificial	1.920059	1.1735880000	3.1413290000	0.0093976790	Very Strong	92.0059
EnvLightingCondition_Dawn__artificial	1.421343	0.7331952000	2.7553570000	0.2978460000	Little Evidence	42.1343
EnvLightingCondition_Daylight__artificial	1.411472	0.8141468000	2.4470450000	0.2196078000	Little Evidence	41.1472

EnvLightingCondition_Dusk	1.123387	0.7427058000	1.6991900000	0.5815796000	Little Evidence	12.3387
EnvLightingCondition_Dark__artificial	0.941325	0.8391680000	1.0559190000	0.3022444000	Little Evidence	-5.8675
EnvLightingCondition_Dark	0.890076	0.7931498000	0.9988476000	0.0477523300	Strong	-10.9924
EnvLightingCondition_Dawn	0.849296	0.4892002000	1.4744550000	0.5616625000	Little Evidence	-15.0704

Model A [Light Conditions] (Predictive Probabilities):

EnvironmentLighting	**PredProb_Severe**
EnvLightingCondition_Other	0.999992
EnvLightingCondition_Dusk__artificial	0.846774
EnvLightingCondition_Dawn__artificial	0.803571
EnvLightingCondition_Daylight__artificial	0.802469
EnvLightingCondition_Dusk	0.76378
EnvLightingCondition_Dark__artificial	0.730409

EnvLightingCondition_Dark	0.719245
EnvLightingCondition_Dawn	0.709677

Model B [Road Surface] (Odds Ratio, Confidence Interval, P-values)

Road Surface	Odds Ratio	2.50%	97.50%	pval_new	Consideration	Odds Inc./Dec.
EnvRoadSurfaceCondition_Spilled_liquid	99965490.000000000000000	0	inf	0.999118	Little Evidence	9996548900.000000
Intercept	2.775234000000000	2.64143	2.91582	0	Very Strong	177.523400
EnvRoadSurfaceCondition_Other	2.030951000000000	1.06755	3.86376	0.030837	Strong	103.095100
EnvRoadSurfaceCondition_Packed_Snow	1.297188000000000	0.48102	3.49818	0.607202	Little Evidence	29.718800
EnvRoadSurfaceCondition_Wet	1.109321000000000	0.98166	1.25358	0.096259	Moderate	10.932100

EnvRoadSurfaceCondition_Loose_Sand_or_Gravel	1.080990000000000	0.21802	5.35989	0.924049	Little Evidence	8.099000
EnvRoadSurfaceCondition_Slush	0.888814100000000	0.48683	1.62271	0.70115	Little Evidence	-11.118590
EnvRoadSurfaceCondition_Loose_Snow	0.732283600000000	0.47501	1.12889	0.158253	Little Evidence	-26.771640
EnvRoadSurfaceCondition_Ice	0.417982800000000	0.24427	0.71524	0.001459	Very Strong	-58.201720

Model B [Road Surface] (Predictive Probabilities):

RoadSurface	**PredProb_Severe**
EnvRoadSurfaceCondition_Spilled_liquid	1
EnvRoadSurfaceCondition_Other	0.849315
EnvRoadSurfaceCondition_Packed_Snow	0.782609
EnvRoadSurfaceCondition_Wet	0.754819
EnvRoadSurfaceCondition_Loose_Sand_or_Gravel	0.75
EnvRoadSurfaceCondition_Slush	0.711538

EnvRoadSurfaceCondition_Loose_Snow	0.670213
EnvRoadSurfaceCondition_Ice	0.537037

APPENDIX C

All questions under Time were answered using Appendix C. *Baseline: 2006 (Year), January (Month), 12 (Hour), 12:00–14:59 (Time Range), Peak (Peakoff - Peak), Non - Winter (Season)*

Model A [Peak] (Odds Ratio, Confidence Interval, P-values):

	Odds Ratio	**2.50%**	**97.50%**	**pval_new**	**Consideration**	**Odds Inc./Dec.**
Intercept	3.117012	2.913327	3.334937	0.0000000000	Very Strong	212
TimePeakOffPeak_OFF_PEAK	0.829925	0.758664	0.907879	0.0000470367	Very Strong	-17.0075

Model A [Peak] (Predictive Probabilities):

PeakOffPeak	**PredProb_Severe**
TimePeakOffPeak_OFF_PEAK	0.721207

Model B [Season] (Odds Ratio, Confidence Interval, P-values):

	Odds Ratio	2.50%	97.50%	pval_new	Consideration	Odds Inc./Dec.
Intercept	2.832131	2.694002	2.977342	0	Very Strong	183.2131
TimeSeason_WINTER	0.963552	0.863875	1.074729	0.505143	Little Evidence	-3.6448

Model B [Season] (Predictive Probabilities):

Season	PredProb_Severe
TimeSeason_WINTER	0.731825

Model C [Season] (Odds Ratio, Confidence Interval, P-values):

	Odds Ratio	2.50%	97.50%	pval_new	Confidence	Odds Inc./Dec.
Intercept	2.470588	2.09974	2.906934	0.0000000000	Very Strong	147.0588
TimeMonth_November	1.472693	1.167021	1.858429	0.0011093100	Very Strong	47.2693
TimeMonth_October	1.36113	1.089294	1.700802	0.0066793880	Very Strong	36.113

TimeMonth_December	1.220643	0.969139	1.537416	0.0903271300	Moderate	22.0643
TimeMonth_September	1.206286	0.971779	1.497385	0.0890531100	Moderate	20.6286
TimeMonth_March	1.142173	0.900814	1.448201	0.2724078000	Little Evidence	14.2173
TimeMonth_June	1.123828	0.906373	1.393454	0.2873308000	Little Evidence	12.3828
TimeMonth_February	1.099353	0.862463	1.401309	0.4442763000	Little Evidence	9.9353
TimeMonth_July	1.088187	0.876654	1.350761	0.4434915000	Little Evidence	8.8187
TimeMonth_August	1.078885	0.872147	1.334629	0.4841960000	Little Evidence	7.8885
TimeMonth_May	1.054103	0.844643	1.315507	0.6410932000	Little Evidence	5.4103
TimeMonth_April	0.888636	0.707646	1.115917	0.3095877000	Little Evidence	-11.1364

Model C [Season] (Predictive Probabilities):

Month	**PredProb_Severe**
TimeMonth_November	0.784409
TimeMonth_October	0.770789
TimeMonth_December	0.750978
TimeMonth_September	0.748759
TimeMonth_March	0.738346

TimeMonth_June	0.735206
TimeMonth_February	0.730897
TimeMonth_July	0.728884
TimeMonth_August	0.727184
TimeMonth_May	0.72255
TimeMonth_April	0.687055

Model D [Hour] (Odds Ratio, Confidence Interval, P-values):

	Odds Ratio	**2.50%**	**97.50%**	**pval_new**	**Consideration**	**Odds Inc./Dec.**
Intercept	2.69919	2.19474	3.31958	0.0000000000	Very Strong	169.9187
TimeHour_7	1.43941	1.01508	2.04112	0.0409555700	Strong	43.9413
TimeHour_8	1.36177	0.9791	1.894	0.0665792400	Moderate	36.1771
TimeHour_18	1.328	1.00547	1.75397	0.0456710400	Strong	32.7995
TimeHour_9	1.25714	0.91824	1.72113	0.1533607000	Little Evidence	25.7141
TimeHour_15	1.2493	0.93918	1.66182	0.1262739000	Little Evidence	24.93
TimeHour_6	1.23832	0.88722	1.72837	0.2089163000	Little Evidence	23.8323
TimeHour_19	1.23207	0.92603	1.63925	0.1520097000	Little Evidence	23.2068

TimeHour_11	1.17683	0.86847	1.59466	0.2935853000	Little Evidence	17.6825
TimeHour_17	1.1473	0.87151	1.51035	0.3272942000	Little Evidence	14.7299
TimeHour_10	1.07999	0.79812	1.46141	0.6180093000	Little Evidence	7.999
TimeHour_22	1.05762	0.78728	1.42078	0.7099433000	Little Evidence	5.7615
TimeHour_21	1.05688	0.7923	1.4098	0.7066905000	Little Evidence	5.6879
TimeHour_4	0.98369	0.61205	1.58101	0.9458570000	Little Evidence	-1.6307
TimeHour_14	0.94679	0.71642	1.25123	0.7006855000	Little Evidence	-5.3213
TimeHour_1	0.91158	0.65048	1.2775	0.5908181000	Little Evidence	-8.8419
TimeHour_16	0.91058	0.69199	1.19823	0.5036315000	Little Evidence	-8.9418
TimeHour_20	0.90263	0.68559	1.18837	0.4653516000	Little Evidence	-9.7371
TimeHour_5	0.87569	0.60059	1.27678	0.4902071000	Little Evidence	-12.4315
TimeHour_2	0.85884	0.62086	1.18805	0.3580184000	Little Evidence	-14.1156
TimeHour_13	0.84291	0.63534	1.11829	0.2360730000	Little Evidence	-15.7091
TimeHour_0	0.81439	0.60018	1.10506	0.1873525000	Little Evidence	-18.5607
TimeHour_23	0.81267	0.60416	1.09314	0.1702994000	Little Evidence	-18.733
TimeHour_3	0.72875	0.52585	1.00993	0.0573542700	Moderate	-27.125

Model D [Hour] (Predictive Probabilities):

Hour	PredProb_Severe
TimeHour_7	0.795302
TimeHour_8	0.786127
TimeHour_18	0.781874
TimeHour_9	0.772379
TimeHour_15	0.771277
TimeHour_6	0.769716
TimeHour_19	0.768817
TimeHour_11	0.760563
TimeHour_17	0.755906
TimeHour_10	0.744578
TimeHour_22	0.740576
TimeHour_21	0.740443
TimeHour_4	0.726415
TimeHour_14	0.71875
TimeHour_1	0.711027
TimeHour_16	0.710801
TimeHour_20	0.708995

TimeHour_5	0.702703
TimeHour_2	0.69863
TimeHour_13	0.694672
TimeHour_0	0.687324
TimeHour_23	0.686869
TimeHour_3	0.662963

Model E [Time Range] (Odds Ratio, Confidence Interval, P-values):

	Odds Ratio	**2.50%**	**97.50%**	**pval_new**	**Consideration**	**Odds Inc./Dec.**
Intercept	2.49882	2.23295	2.79636	0.0000000000	Little Evidence	149.8824
TimeRange_6_00_to_8_59	1.44876	1.19767	1.75249	0.0001348602	Very Strong	44.8759
TimeRange_9_00_to_11_59	1.25985	1.06042	1.4968	0.0086108970	Very Strong	25.9852
TimeRange_18_00_to_20_59	1.22994	1.05229	1.43757	0.0093117180	Very Strong	22.9937
TimeRange_15_00_to_17_59	1.17655	1.00739	1.37413	0.0400824500	Very Strong	17.6554
TimeRange_21_00_to_23_59	1.05347	0.89389	1.24154	0.5342633000	Strong	5.3469

TimeRange_Midnight_to_2_59	0.92407	0.77126	1.10716	0.3918764000	Little Evidence	-7.5929
TimeRange_3_00_to_5_59	0.8827	0.71472	1.09016	0.2466716000	Little Evidence	-11.7299

Model E [Time Range] (Predictive Probabilities):

TimeRange	**PredProb_Severe**
TimeRange_6_00_to_8_59	0.783559
TimeRange_9_00_to_11_59	0.758929
TimeRange_18_00_to_20_59	0.754505
TimeRange_15_00_to_17_59	0.746193
TimeRange_21_00_to_23_59	0.724702
TimeRange_Midnight_to_2_59	0.697802
TimeRange_3_00_to_5_59	0.688057

APPENDIX D

All questions were answered using Appendix D

- **P-values** (Answer the question: Is the effect of the variable real or is it a random chance?)
    - ***Interpretation***

      <0.01 <0.05 <0.1 >0.1

      Very Strong | Strong | Moderate | Little Evidence

We will be disregarding the: *Moderate, Little Evidence*, and considering: *Very Strong, Strong*

- **Confidence Interval** (Answer the question: Is the effect statistically significant?)
    - ***Interpretation***
        - *[$CI_{Lower} > 1$, $CI_{Upper} > 1$]: Significant Positive Impact*
        - *[$CI_{Lower} < 1$, $CI_{Upper} < 1$]: Significant Negative Impact*
        - *[$CI_{Lower} < 1$, $CI_{Upper} > 1$]: Not Statistically Significant*

- **Odds Ratio (OR) (***exp(coefficient)***)**
    - ***Context Interpretation***
        - *<X> has <y times> higher odds of <outcome> than <baseline>, Example: cyclist have 34 x higher odds of a severe injury than drivers*

- ***Odds Increase/Decrease***

- ***Context Interpretation***
    - *X has Y% higher/lower odds of <outcome>than <baseline>*
- ***Formula:*** *(Odds Ratio -1) * 100 %*

- ***Predictive Probability***
    - ***Context Interpretation***
        - *In this context, the % chance of a severe injury*

- ***Percentage Points***
    - ***Interpretation***
        - *The difference between two percentage points*

- ***Predictive Probability of Intercept***
    - ***Context Interpretation:*** *On some occasions we compare Winter to non- winter (or peak to Off-peak) and the predictive probability for the baseline is not computed.*
        - *Formula*
            - *Predictive Probability* $_{Baseline}$ *= Odds Ratio / (1 + Odds Ratio)*
- ***Intercept vs. Baseline***

- *Intercept: Outcome when all predictors = 0 (used for computing **baseline predictive probabilities**)*
- *Baseline: For group comparison (baseline = 1 for plotting categorical odds ratio)*

Citations

- Hosmer, D. W., Lemeshow, S., & Sturdivant, R. X. (2013). *Applied Logistic Regression* (3rd ed.). Wiley.
- Kleinbaum, D. G., & Klein, M. (2010). *Logistic Regression: A Self-Learning Text* (3rd ed.). Springer.
- UCLA IDRE. (n.d.). *FAQ: How do I interpret odds ratios in logistic regression?*